MILITARY SPECIFICATION

WAX, BONE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, bone wax suitable for use in orthopedic surgery and suitable for hemostasis in the treatment of bond cavities.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Set-Up.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific proecurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.
Pharmacopeia of the United States (U.S.P.)

(Application for copies should be addressed to the Mack publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Bone wax shall be a mixture of bees wax and almond oil, with 2 percent (± 0.5 percent) salicylic acid or a mixture of mismo bees wax and isopropyl palmitate. The ratio of ingredients shall be such that the melting point of the bone wax shall not be less than 57.0 degrees Centigrade (° C.) nor more than 62.8° C. The finished product shall be homogeneous, and of such consistency as to be suitable for hemostasis in the treatment of bone cavities. All ingredients used in the bone wax shall comply with the current edition of the Pharmacopeia of the United States.

3.2 Packet assembly. Each packet shall consist of 2 to 3 grams (gm.) of bone wax. Each packet shall be packed in an inner and outer container as specified in 3.2.1 and 3.2.2, or, as an alternate, each packet shall be packed in tube type containers as specified in 3.2.3, 3.2.4 and 3.2.5.

3.2.1 Inner container. The foil container shall be a combination of plastic and aluminum foil closed by heat sealing, and shall be fabricated in such a manner and of such material as to maintain the sterility of the bone wax. The sealed foil container shall have two free tabs along the top edge capable of being readily peeled back along the edges to expose the bone wax.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

3.2.2 Outer container. The outer container shall be fabricated of a combination of plastic and aluminum foil closed by heat sealing, and shall be fabricated in such a manner and of such material as to maintain the sterility of the inner container. The sealed outer container shall have two free tabs along the top edge, capable of being readily peeled back along the sealed edges to expose the inner container. The outer container shall be so designed as to permit ready dispensing of the sterile inner container by aseptic techniques.

3.2.3 Dispenser. The sterile bone wax shall be contained in a suitable tin or tin-copper alloy tube dispenser with an open end. The weight of the wax and dispenser shall not exceed 6 grams. The dispenser shall be suitable for dispensing bone wax during surgical procedures.

3.2.4 Inner container. The sterile inner container shall be fabricated from clear, clean flint glass tubing with one open end sealed with a suitable polyethylene plastic cap. It shall be fabricated in such a manner as to maintain sterility of the dispenser. It shall also allow for ready dispensing of the sterile dispenser by aseptic techniques.

3.2.5 Outer container. The outer container shall be fabricated from cellulose acetate butyrate plastic tubing with one open end sealed with a suitable polyethylene plastic cap. Outer container dimensions shall not exceed 13/16 inch outer diameter and 4-1/4 inches in length. It shall be fabricated in such a manner as to maintain sterility of the inner container. It shall also allow for ready dispensing of the inner container by aseptic techniques.

3.3 Sterility. The entire contents of the packet assembly shall be sterile when tested as specified in 4.4.1.

3.4 Salicylic acid content determination. The salicylic acid content shall be determined as specified in 4.4.2.

3.5 Melting point. The melting point of the wax shall be no less than 57° C. nor more than 62.8° C. when tested as specified in 4.4.3.

3.6 Item identification. The inner container shall be legibly marked with the item name, and name or registered trademark of the manufacturer.

3.7 Lot (control) numbers. The contractor shall maintain lot (control) numbers in order to identify the sterilization of each lot. The lot (control) numbers shall be entered in the contractor's original and permanent records of sterilization and shall also appear on each shipment as hereinafter specified. The manufacturer's records shall be open for Government inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

3.8 Certificate of sterility. When inspection is made at destination, a certificate of sterility, listing all lot or control numbers, stating that samples from each sterility lot have been tested in accordance with the U.S.P. method and have been found to be sterile, shall be submitted by the contractor to the Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, Attention: Quality Assurance Branch. A copy of the certificate shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the cognizant Government inspector.

3.9 Workmanship. Workmanship shall be first class throughout. Bone wax shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination purposes shall be one bone wax.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)*
Major	I	1.0
Minor	I	2.0

* If acceptance number is not exceeded, lot shall be acceptable. If rejection number is equalled or exceeded, inspection at Level II may be performed to determine final acceptance or rejection.

4.2.2 Sampling for test. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. Where possible, the same sample shall be used for the determination of two or more test characteristics.

Table II. Sampling for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
1-Bone Wax	Sterility	3.3	4.4.1	**	*
1-Bone Wax	Salicylic acid	3.4	4.4.2	***	*
1-Bone Wax	Melting point	3.5	4.4.3	***	*

* Acceptance number shall be zero for all sample sizes.
** Sampling shall be as indicated in the U.S.P. for Crystalline or Powdered Solids under Sterility Tests, in the U.S.P.
*** A minimum of one sample from each lot of bone wax shall be tested.

4.3 Examination. The bone wax shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include but shall not be limited to the defects listed in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

Table III. Classification of Defects

Categories	Defects
Major	
101	Material not as specified.
102	Melting point not as specified.
103	Packet assembly not as specified.
104	Bone wax not homogeneous.
Minor	
201	Packaging and packing not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Sterility test. The packets shall be tested for sterility in accordance with the current edition of the U.S.P.

4.4.2 Method for determination of salicylic acid in bone wax. The determination of salicylic acid for compliance with requirements of this specification shall be performed as hereinafter specified.

4.4.2.1 Reagents. Reagents to be used in determination of salicylic acid shall be as follows:

(1) Benzene-Methanol Solution: Benzene, U.S.P. Reagent, 3 Volumes, plus Methanol, U.S.P. Reagent, 1 Volume.

(2) Ferric nitrate solution: 1.0 gm. $Fe(NO_3)_3 \cdot 9H_2O$ in 100 cubic centimeters (cc.) of 1.0% HNO_3.

(3) Standard solutions. Accurately weigh exactly 0.500 gm. of salicylic acid and dissolve in water to make 500 cc. Four standard solutions shall be prepared by diluting each of 0.5 cc., 1.0 cc., 1.5 cc., and 2.0 cc. portions of the salicylic acid solution, described above, with distilled water to make 50 cc.

4.4.2.2 Procedure. Dissolve approximately 50 milligrams (mg.) of bone wax, accurately weighed, in 3 cc. of Benzene-Methanol Solution contained in a 50 cc. volumetric flask, warming gently, if necessary. To the cooled solution add exactly 5.0 cc. of ferric nitrate solution. Then add enough water to make approximately 45 cc. and shake vigorously for 1 minute. Allow the two immiscible liquids to separate, and then add water to bring the aqueous layer to the 50 cc. mark. Draw off the benzene

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

4.4.2.2 (cont'd.)

layer by means of a medicine dropper, pipette, or other suitable means. Centrifuge a portion of the aqueous solution to free the solution from suspended benzene droplets. Determine the optical density of this aqueous solution by use of a Beckman Model DU spectrophotometer, at wave length 525 millimicrons and using a 1-centimeter corex cell and tungsten lamp. Use a solution, prepared with the same quantities of the same reagents in a similar manner as a blank. Prepare a calibration curve of the standard solutions (prepared as specified in 4.4.2.1 [3]) employing the above method and reagents. Determine the concentration of the sample solution from this curve.

4.4.3 Method for determination of melting point. The melting point determination shall be conducted in accordance with procedure for class II or III as specified, under "Melting Range or Temperature", in the current edition of the Pharmacopeia of the United States. Three samples from each lot shall be tested. If the variation among the three determinations is less than 1 degree, their arithmetical average shall be considered as the melting point of the lot represented. If the variation among the three determinations is greater than 1 degree, two additional determinations shall be made and the arithmetical average of the five determinations shall be considered as the melting point of the lot represented.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing twelve packets of wax, as specified, constitute one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit packaging.

5.1.3.1.1 Immediate container (Packet). Each immediate container (packet) shall be a combination of an inner and an outer container as specified in 3.2.1 and 3.2.2.

5.1.3.1.2 Unit package. Twelve (12) immediate containers (packets) shall be overpackaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable. Box closure shall be adequate to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size, constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking .

5.3.1 Inner container (packet). Each inner container (packet) shall be marked as specified in 3.6.

5.3.2 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water resistant material. Contract or purchase order number and name of contractor shall be shown. In addition, lot (control) number shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, lot (control) number shall be shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The wax bone is intended for use in orthopedic surgery and suitable for hemostasis in the treatment of bone cavities.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6515-965-2511	WAX, BONE, 12 Packets

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

6.4 Supersession data. This specification includes requirements of Purchase Description No. 3, dated 1 August 1967.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1608

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the [illegible] Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to [illegible]ity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion [illegible] document (s) or serve to amend contractual requirements.

SPECIFICATION NUMBER: MIL-P-25725

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.

CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE [illegible] TED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MA, RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

Fold

DPSC FORM 2418
MAR 69

EDITION AUG 66, WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:20Z
Check the source to verify that this is the current version before use.